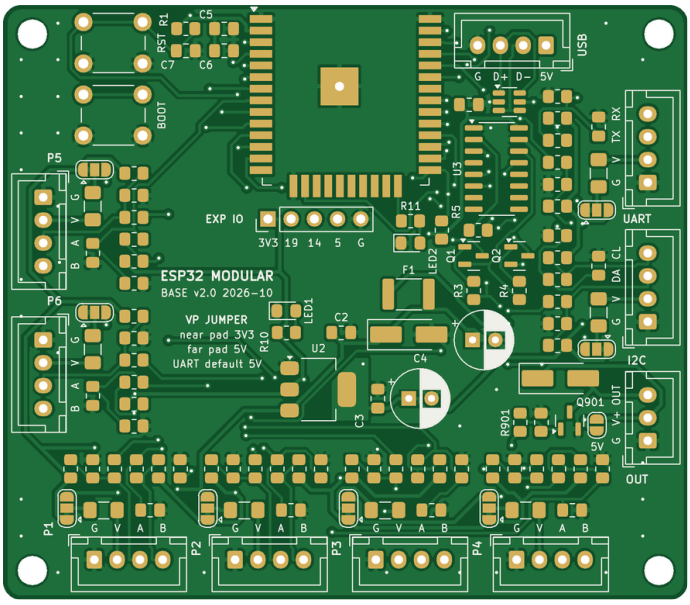
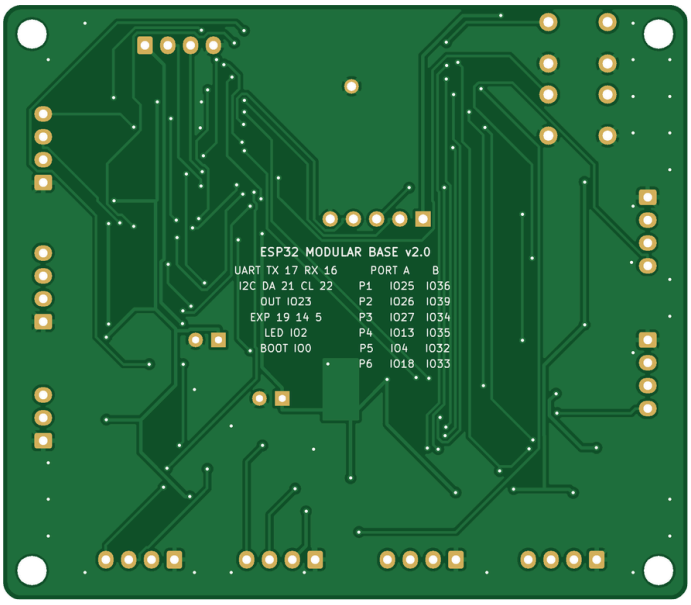
ESP32 Modular Base v2.0: assembly guide

Hand assembly · rev 2.0 · 2026-10



Top



Bottom

BOARD	PARTS TO SOLDER	LEFT EMPTY	BATCH
75 × 65 mm, 2 layers, 1.6 mm	68 per board	38 option footprints	5 boards (tick columns B1–B5)

How to use this guide

1. Work through the steps in order. Each step is one page: a board drawing with only that step's parts in colour, and a table.
2. Do one step on **all 5 boards** before moving to the next. Tick the box for each board (B1–B5) as you finish a line.
3. Number the boards on the back with a marker so the ticks and test results stay with the right board.
4. Colours in a step = values. Black dots mark pin 1 (**1**), cathode (**K**) or plus (**+**).
5. Footprints drawn dashed stay **empty**. They are options for configuring ports later.
6. Grid squares (A–H, 1–7) help find a part: the parts index on the last page lists every reference.

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Step 2: Transistors and 3.3 V regulator	4 parts per board	**5**
Step 3: Core 0805 resistors and capacitors	15 parts per board	**6**
Step 4: Port parts fitted by default	27 parts per board	**7**
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Before you start

Tools

Soldering iron	Temperature controlled. Chisel or hoof tip 1.6–2.4 mm for drag soldering, a fine conical tip for 0805, a big chisel for the regulator tab and THT ground pins.
Solder	0.5–0.8 mm wire with flux core. Leaded 63/37 is easiest; lead-free SAC305 needs about 30 °C more.
Flux	No-clean flux pen or gel. Use plenty on the CH340C, USBLC6 and the module.
Desoldering wick	1.5–2 mm, for bridges.
Tweezers	Fine straight or curved, non-magnetic if possible.
Magnification	10× loupe or a USB microscope for checking IC pins and the module.
Cleaning	Isopropyl alcohol (99 %) and an old toothbrush.
Multimeter	Continuity, diode test, DC volts.
Other	Kapton tape, side cutters, a USB-A to C cable, a PC with a USB port.

Iron temperature (guide)

0805, SOT-23	320–340 °C leaded · 350–370 °C lead-free
SOIC, module	330–360 °C leaded · 360–380 °C lead-free
U2 tab, F1, THT GND pins	360–390 °C with a large tip, hold longer

Prepare the boards

1. Check each bare PCB: no scratches across tracks, holes clear, edges clean.
2. Number the boards 1–5 on the back (marker, near a corner).
3. Continuity check on one bare board: **+5V to GND** and **+3V3 to GND** must be open (no beep). Probe +5V on the C1 square pad, +3V3 on the C4 square pad, GND on any port pin **G**.
4. Sort the parts by step. Keep each value in its own labelled bag; unmarked 0805 capacitors look identical.
5. Work on an ESD mat or touch a grounded object before handling the CH340C, USBLC6 and the module.

Symbols on the drawings

part of this step (colour = value)

leave empty (option)

already fitted or later step

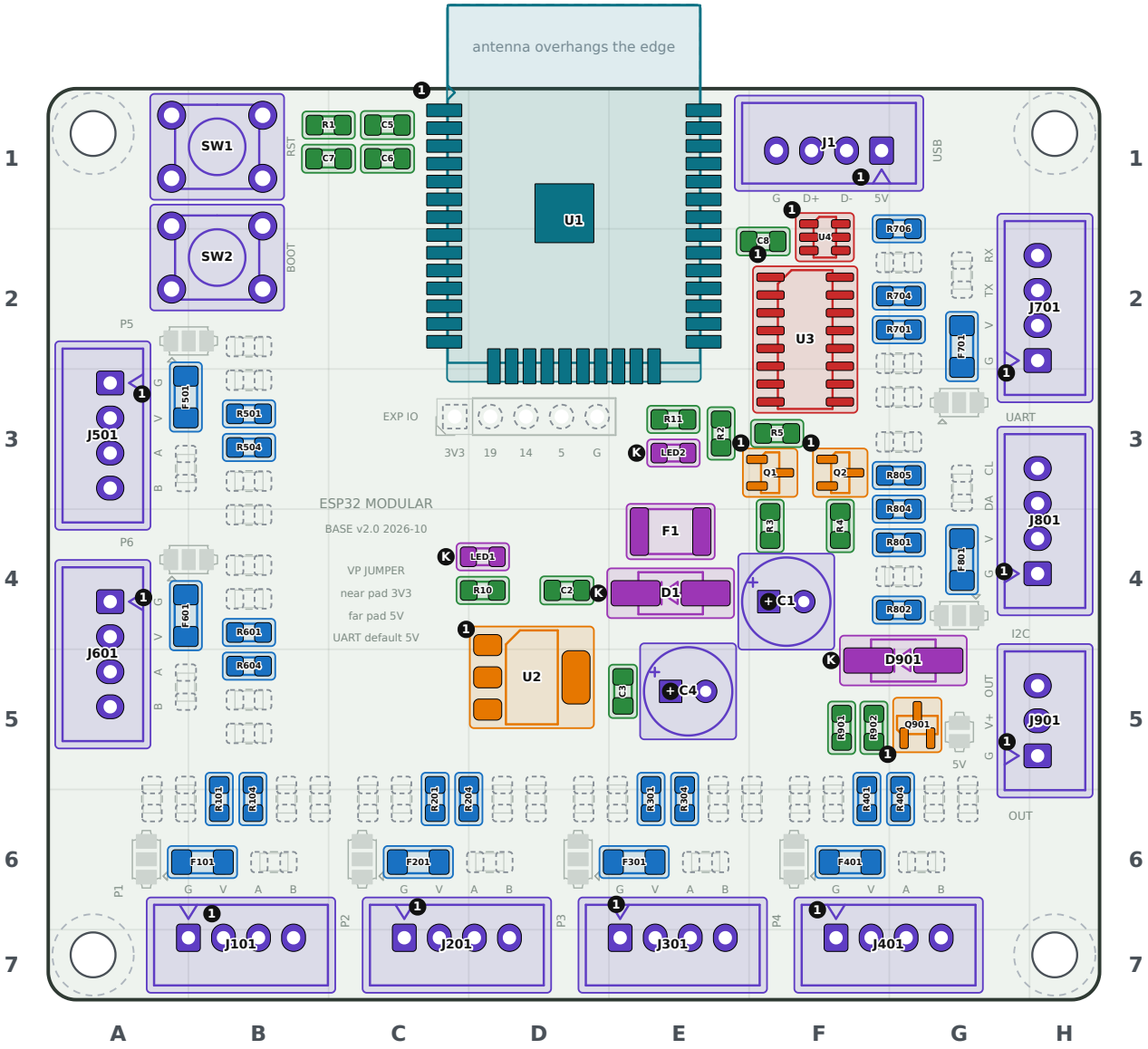
1 pin 1 · K cathode · + plus

Good joint, bad joint

- **Good:** shiny (leaded) or satin (lead-free), concave fillet from pad to part, pad fully wetted.
- **Cold:** grey, lumpy, ball sitting on the pad. Reheat with flux.
- **Bridge:** solder joining two pins. Flux + drag, or wick.
- **Tombstone:** 0805 standing on one end. Reheat the high end and press down.

Board map

All fitted parts, coloured by assembly step



Step 1 Fine-pitch ICs Step 2 Transistors and 3.3 V regulator Step 3 Core 0805 resistors and capacitors

Step 4 Port parts fitted by default Step 5 Diodes, LEDs and input PTC Step 6 ESP-32S module

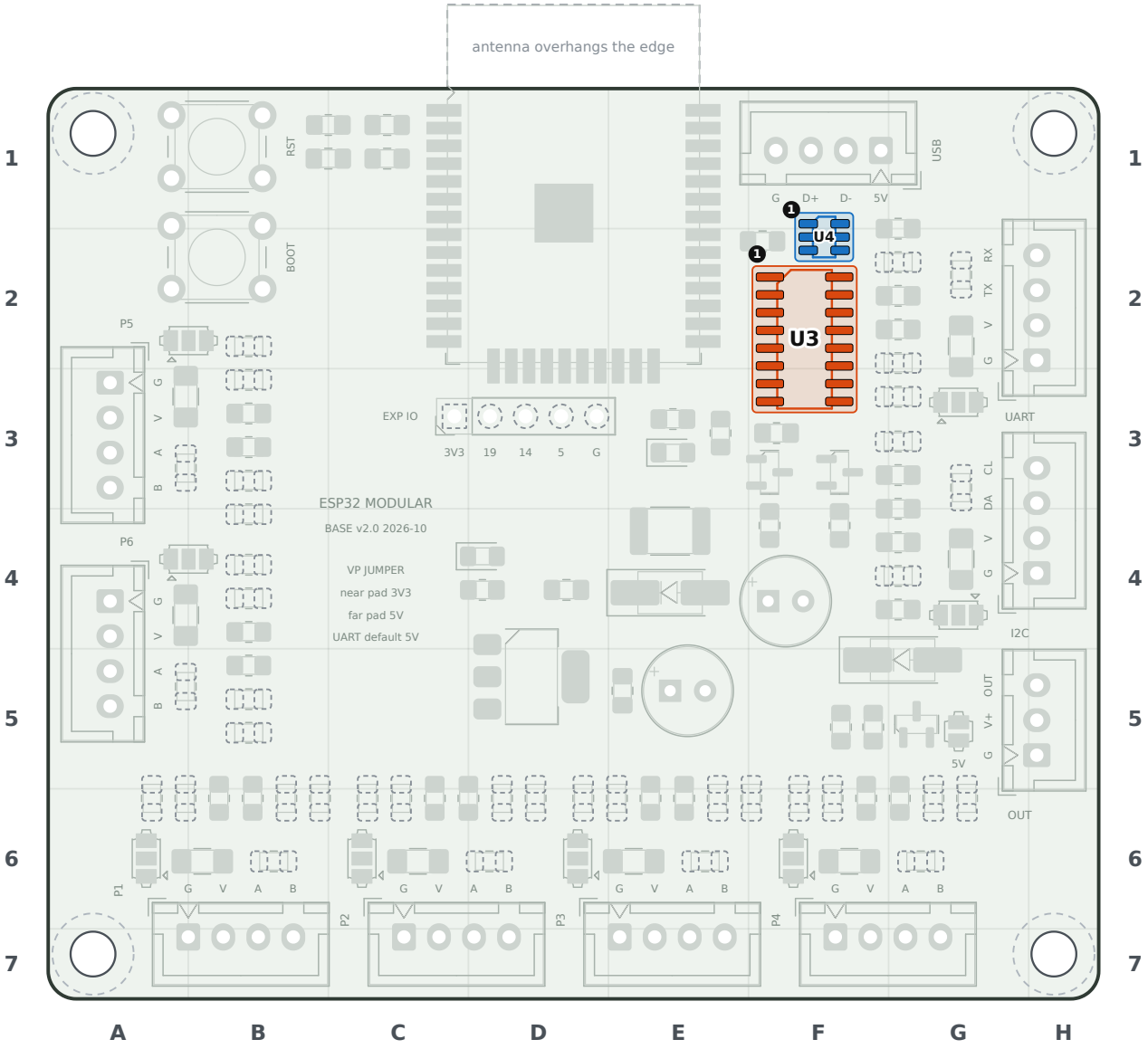
Step 7 Through-hole parts

CONNECTOR	WHERE	PINS (PIN 1 = SQUARE PAD)
J1 USB in	top right	5V, D−, D+, G
J101–J401 P1–P4	bottom	G, VP, A, B
J501, J601 P5, P6	left	G, VP, A, B
J701 UART, J801 I2C	right	G, VP, TX, RX / G, VP, SDA, SCL
J901 OUT	right, low	G, V+, OUT
SW1 RESET, SW2 BOOT	top left	–

Step 1: Fine-pitch ICs

2 parts per board · 10 for 5 boards

Solder these first, while the board is flat and nothing is in the way of the iron.



	QTY	VALUE	REFERENCES	PACKAGE	GRID	ORIENTATION / NOTE	B1	B2	B3	B4	B5
■	1	**CH340C**	U3	SOIC-16	F2	Pin-1 dot/notch on the ●1 marker (top-left)	☐	☐	☐	☐	☐
■	1	**USBLC6-2SC6**	U4	SOT-23-6	F2	Pin-1 dot on the ●1 marker	☐	☐	☐	☐	☐

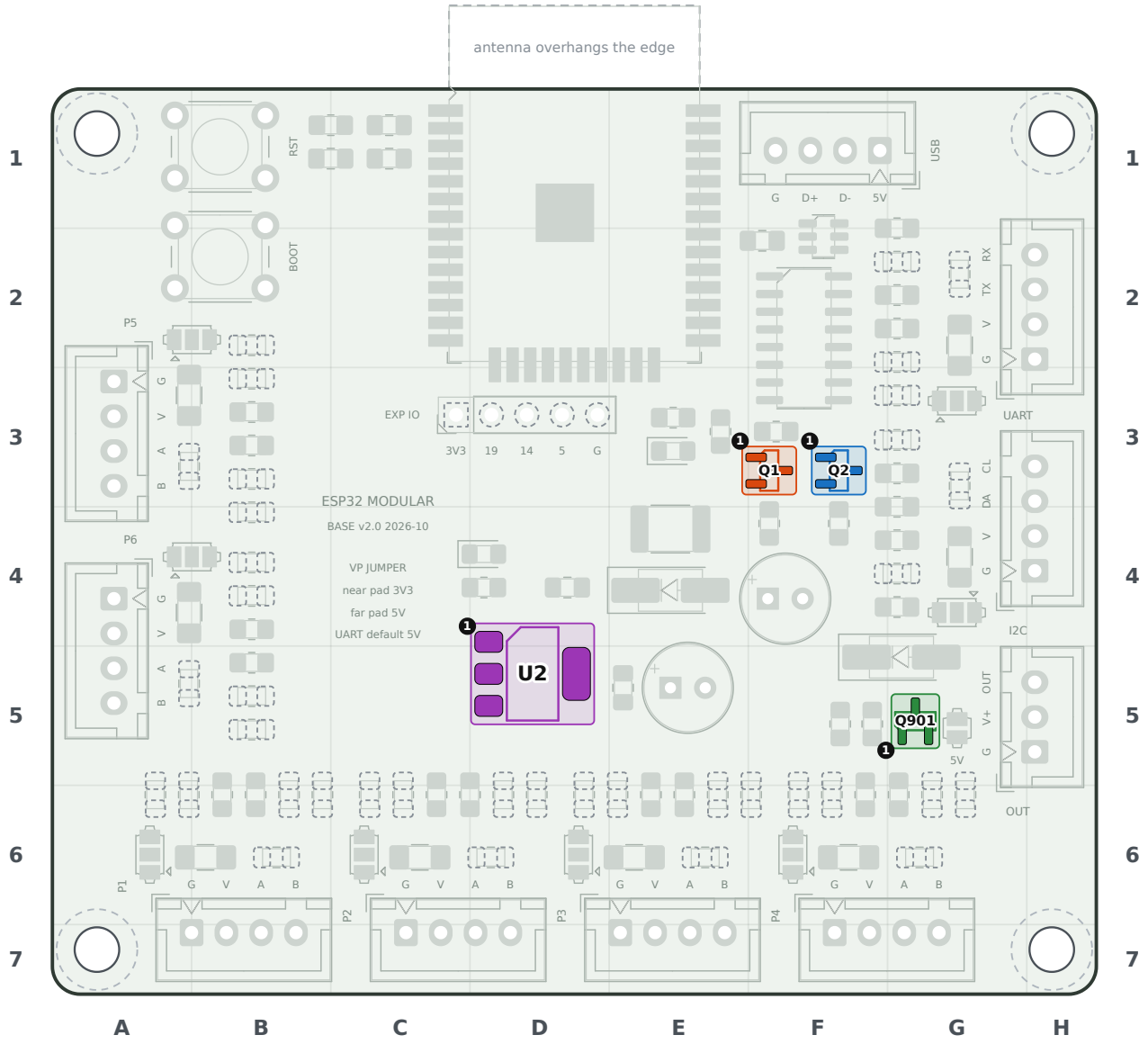
Iron: 330–350 °C leaded, 360–380 °C lead-free

- Flux the pads. Tin **one corner pad** only.
- Place the IC with tweezers, pin 1 on the **●1** marker. Reheat the tinned pad so the pin sinks in.
- Check alignment on all sides with a loupe. If it is off, reheat and nudge; then tack the opposite corner.
- Drag-solder each row: plenty of flux, a little solder on a chisel or hoof tip, pull slowly along the pins.
- Bridges: add flux and drag again, or lift with wick. Check every gap against light.

Step 2: Transistors and 3.3 V regulator

4 parts per board · 20 for 5 boards

SOT-23 and SOT-223 only fit one way round, but they are small and easy to lose: do them before the 0805 parts crowd the area.



	QTY	VALUE	REFERENCES	PACKAGE	GRID	ORIENTATION / NOTE	B1	B2	B3	B4	B5
	1	**MMBT3904**	Q1	SOT-23	F3	Fits one way (3 pads)	☐	☐	☐	☐	☐
	1	**MMBT3904**	Q2	SOT-23	F3	Fits one way (3 pads)	☐	☐	☐	☐	☐
	1	**AO3400A**	Q901	SOT-23	G5	Fits one way (3 pads)	☐	☐	☐	☐	☐
	1	**AMS1117-3.3**	U2	SOT-223	D5	Big tab = right side	☐	☐	☐	☐	☐

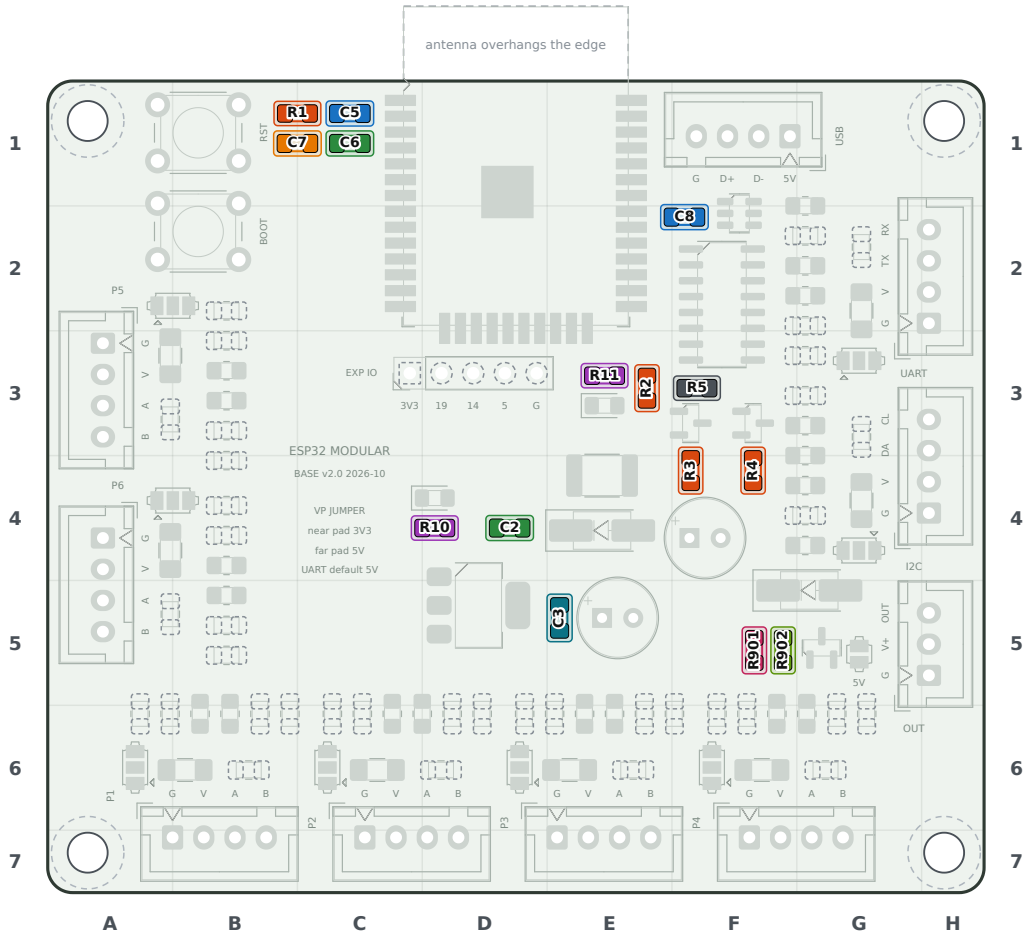
Iron: SOT-23: 330–350 °C. U2 tab: 360–390 °C, larger tip

- Tin one pad, place the part, reheat to fix it, then solder the other pads.
- AMS1117 (U2): the big tab sits on a copper heat spreader. Use a larger tip and more heat, hold 3–5 s until the solder flows over the whole tab.
- Q1/Q2 are MMBT3904 (marking 1AM). Q901 is AO3400A (marking A09T). Do not mix them up.

Step 3: Core 0805 resistors and capacitors

15 parts per board · 75 for 5 boards

Decoupling, reset, auto-reset and LED resistors around the ESP32, CH340C and regulator.



	QTY	VALUE	REFERENCES	PACKAGE	GRID	ORIENTATION / NOTE	B1	B2	B3	B4	B5
	4	**10 k**	R1 R2 R3 R4	0805	C1 E3 F4	EN pull-up, IO0 pull-up, auto-reset bases	☐	☐	☐	☐	☐
	2	**100 nF**	C5 C8	0805	C1 F2	Decoupling	☐	☐	☐	☐	☐
	2	**10 uF**	C2 C6	0805	C1 D4		☐	☐	☐	☐	☐
	2	**1 k**	R10 R11	0805	D4 E3	LED resistors (R10 power, R11 status)	☐	☐	☐	☐	☐
	1	**1 uF**	C7	0805	C1	EN delay	☐	☐	☐	☐	☐
	1	**22 uF**	C3	0805	E5	Regulator output	☐	☐	☐	☐	☐
	1	**100 R**	R901	0805	F5	OUT gate resistor	☐	☐	☐	☐	☐
	1	**100 k**	R902	0805	F5	OUT gate pull-down	☐	☐	☐	☐	☐
	1	**470 R**	R5	0805	F3	CH340 TX to ESP RX	☐	☐	☐	☐	☐

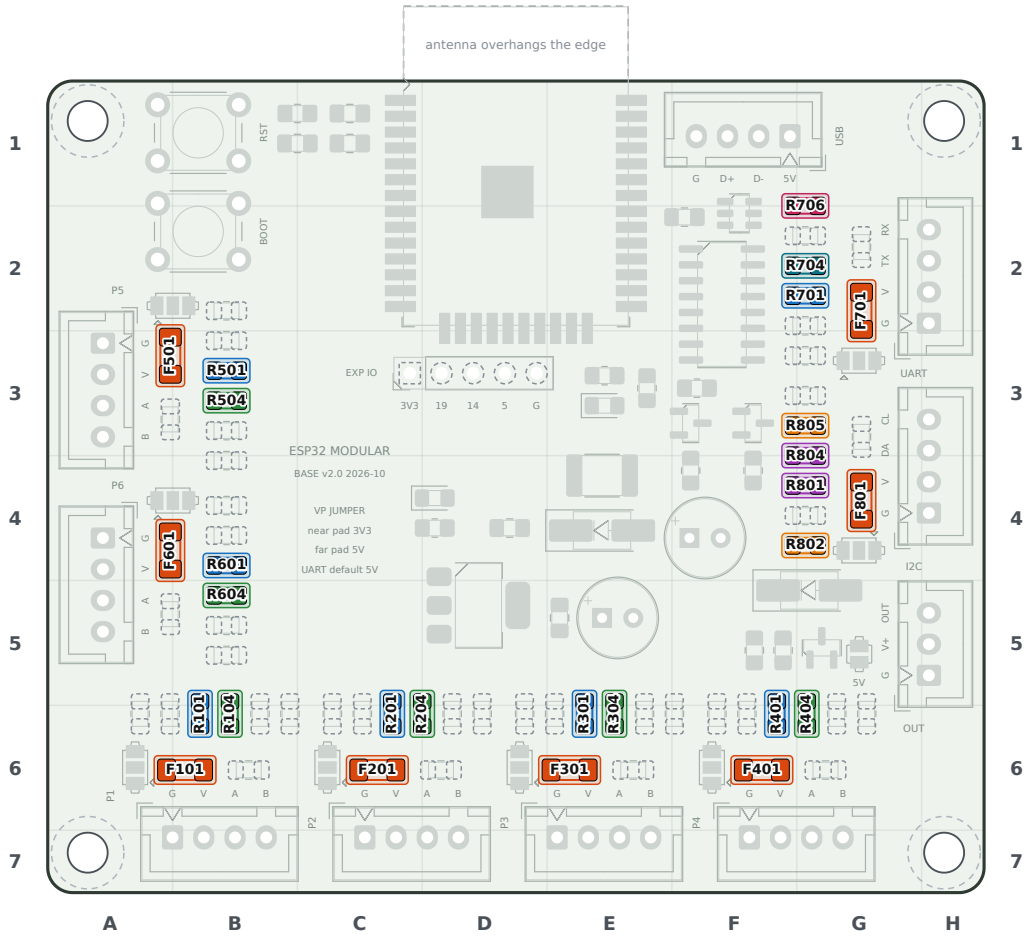
Iron: 320–340 °C leaded, 350–370 °C lead-free

- Work one value at a time: open one bag, fit every position of that value on all boards, close the bag.
- Tin one pad, slide the part in with tweezers, reheat; then solder the second pad.
- Ceramic capacitors have no marking: keep them in their bags until placed.
- Look for tombstoning (part standing up) and a clean fillet on both ends.

Step 4: Port parts fitted by default

27 parts per board · 135 for 5 boards

Every port gets its 330 R (A series), 1 k (B series) and 0.5 A PTC. UART and I2C have their own default values. All other port footprints stay empty (dashed on the diagram) until you configure the port.



	QTY	VALUE	REFERENCES	PACKAGE	GRID	ORIENTATION / NOTE	B1	B2	B3	B4	B5
	8	**PTC 0.5 A hold**	F101 F201 F301 F401 F501 F601 F701 F801	1206	A3 A4 B6 C6 E6 F6 G2 G4	One per port, no polarity	☐	☐	☐	☐	☐
	7	**330 R**	R101 R201 R301 R401 R501 R601 R701	0805	B3 B4 B6 C6 E6 F6 G2	A-pin series resistors + UART TX	☐	☐	☐	☐	☐
	6	**1 k**	R104 R204 R304 R404 R504 R604	0805	B3 B5 B6 D6 E6 G6	B-pin series resistors	☐	☐	☐	☐	☐
	2	**100 R**	R801 R804	0805	G4	I2C series	☐	☐	☐	☐	☐
	2	**4.7 k**	R802 R805	0805	G3 G4	I2C pull-ups	☐	☐	☐	☐	☐
	1	**10 k**	R704	0805	G2	UART RX series (with R706)	☐	☐	☐	☐	☐
	1	**22 k**	R706	0805	G2	UART RX divider bottom	☐	☐	☐	☐	☐

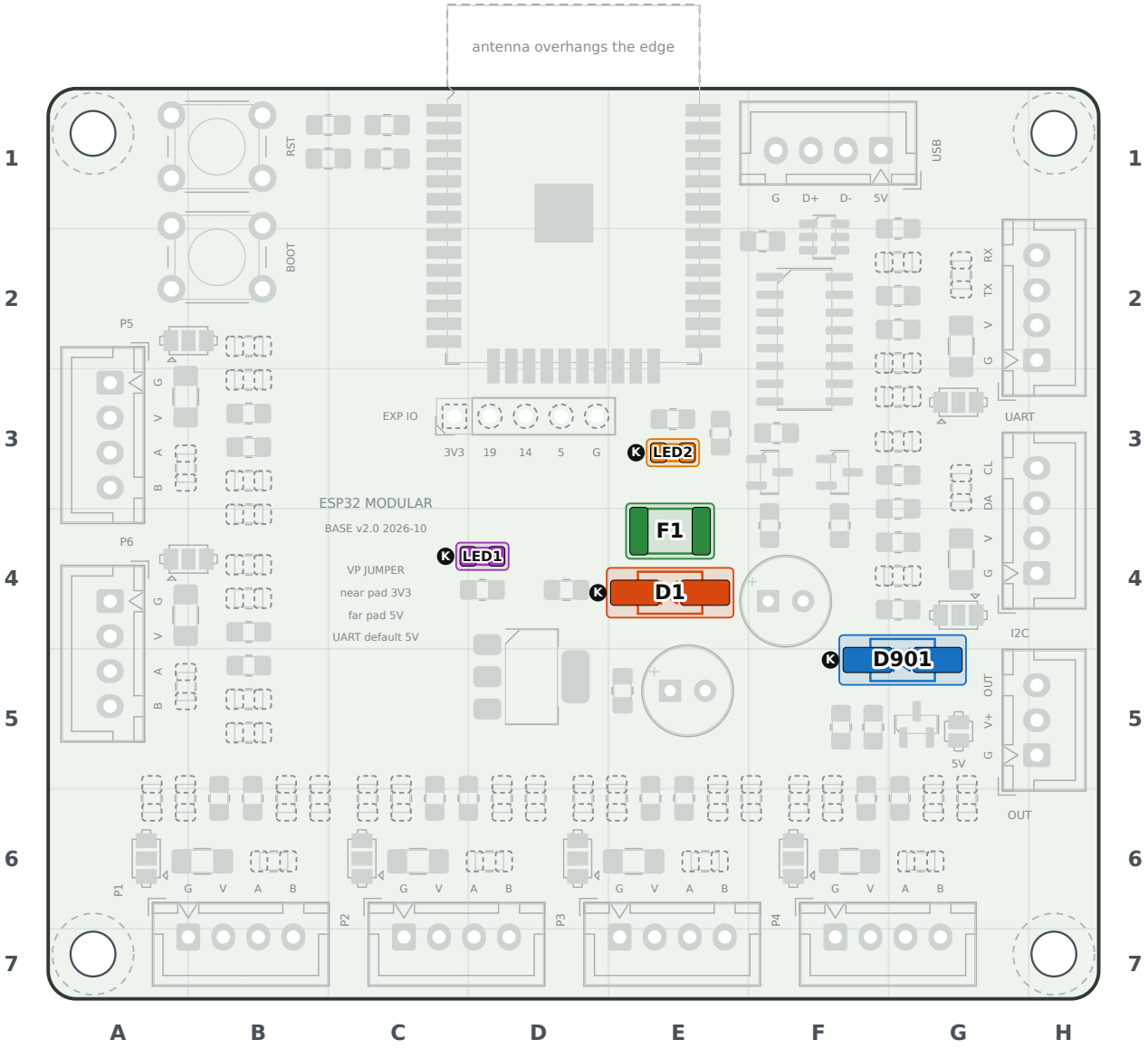
Iron: 320–340 °C leaded

- Colour = value. Do one colour at a time across all boards.
- UART: R704 is 10 k (not 1 k) and R706 is 22 k: together they divide 5 V down for RX.
- I2C: R801 and R804 are 100 R, R802 and R805 are 4.7 k pull-ups.
- PTCs (1206) have no polarity. They need a little more heat than 0805.
- Do **not** fill the dashed footprints now.

Step 5: Diodes, LEDs and input PTC

5 parts per board · 25 for 5 boards

Polarised parts: check every one against the K marker before soldering the second pad.



QTY	VALUE	REFERENCES	PACKAGE	GRID	ORIENTATION / NOTE	B1	B2	B3	B4	B5
1	**SMAJ5.0A**	D1	SMA	E4	Band on K (toward +5V)	☐	☐	☐	☐	☐
1	**SS34**	D901	SMA	G5	Band on K (toward V+)	☐	☐	☐	☐	☐
1	**PTC 1.1 A hold**	F1	1812	E4	No polarity	☐	☐	☐	☐	☐
1	**LED red**	LED1	0805 LED	D4	Cathode on K	☐	☐	☐	☐	☐
1	**LED yellow**	LED2	0805 LED	E3	Cathode on K	☐	☐	☐	☐	☐

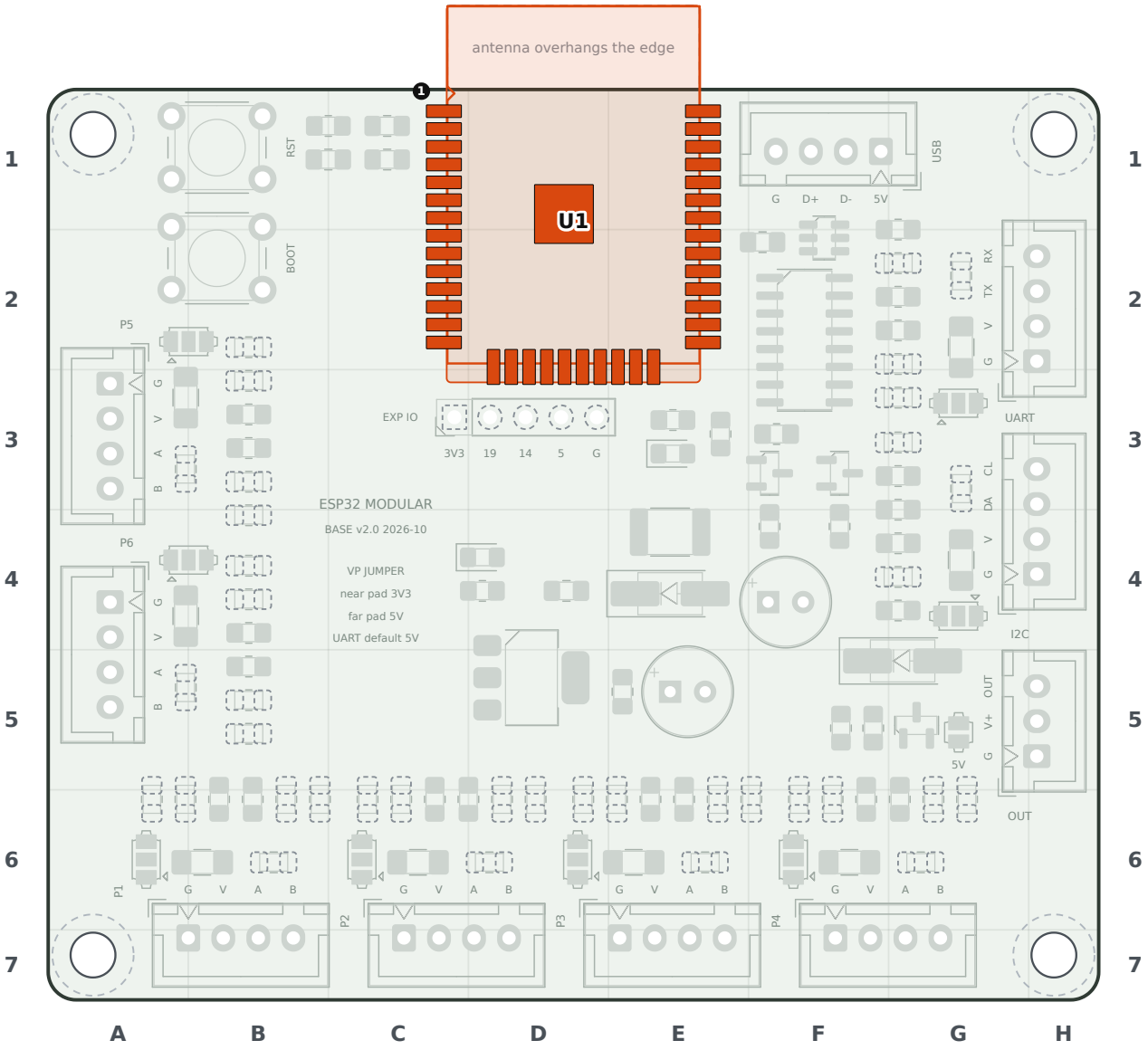
Iron: 340–360 °C, larger tip for F1, D1, D901

- D1 SMAJ5.0A and D901 SS34: the **band** on the body is the cathode and goes on the **K** marker.
- LEDs: the cathode is marked by a green dot/line on top or the bar of the T on the underside. Not sure? Multimeter in diode mode, red probe on the anode: the LED glows.
- LED1 is red (power), LED2 is yellow (status). Do not use blue, green or white here.
- F1 (1812) has a lot of copper: use the bigger tip and give it time.

Step 6: ESP-32S module

1 parts per board · 5 for 5 boards

The module covers nothing else, but once it is on, the area under it is closed. All SMD parts around it are done by now.



	QTY	VALUE	REFERENCES	PACKAGE	GRID	ORIENTATION / NOTE	B1	B2	B3	B4	B5
■	1	**ESP-32S (Ai-Thinker)**	U1	Module	D1	Antenna over top edge, pin 1 top-left	☐	☐	☐	☐	☐

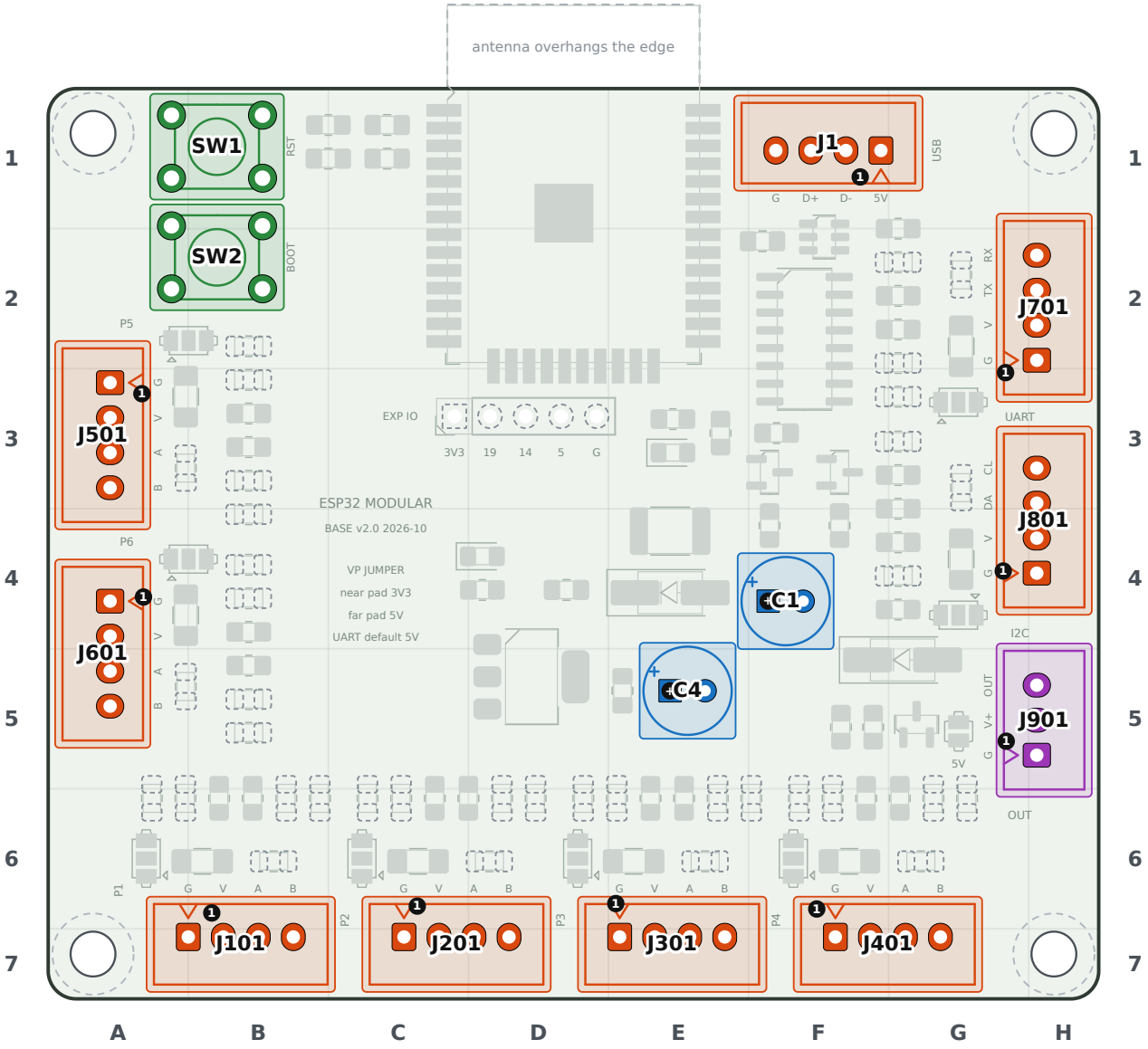
Iron: 340–360 °C, chisel tip

- Antenna end **overhangs the top edge** of the board. Pin 1 is top-left (●1).
- Tin pad 1 only. Place the module, align all 38 castellations with the pads, reheat pad 1.
- Check alignment on the left, right and bottom rows. Tack pad 38 (top-right), check again.
- Drag-solder each side with flux. The pads are extended 1 mm so the iron reaches them.
- Optional: feed solder into the 1 mm hole in the centre GND pad from the back.

Step 7: Through-hole parts

14 parts per board · 70 for 5 boards

Tallest parts last, so the board lies flat for all the SMD work.



	QTY	VALUE	REFERENCES	PACKAGE	GRID	ORIENTATION / NOTE	B1	B2	B3	B4	B5
	9	**XH 4P socket**	J1 J101 J201 J301 J401 J501 J601 J701 J801	XH 4P	A3 A4 B7 C7 E7 F1 F7 H2 H4		☐	☐	☐	☐	☐
	2	**100 uF 10 V**	C1 C4	Radial 6.3 mm	E5 F4		☐	☐	☐	☐	☐
	2	**Tactile switch**	SW1 SW2	6x6 THT	A1		☐	☐	☐	☐	☐
	1	**XH 3P socket**	J901	XH 3P	H5		☐	☐	☐	☐	☐

Iron: 340–360 °C, larger tip for the ground pins

- Insert from the top, solder from the bottom. Hold the part flat with a finger or kapton tape.
- Solder **one pin**, check the part sits flat and square, reheat that pin to correct it, then solder the rest.
- C1 and C4: the long leg is +, the stripe on the can is −. + goes in the square pad (+ marker).
- XH sockets: match the plastic outline to the silkscreen; pin 1 is the square pad.
- Trim leads to about 1 mm after soldering (switch and connector pins are already short).

Inspection

Do this under magnification before any power is applied. Most first-power failures are a bridge on the CH340C or the module, or a part the wrong way round.

Polarity and orientation, part by part

REF	PART	MUST BE	GRID	B1	B2	B3	B4	B5
U3	CH340C	Pin-1 dot/notch on the ●1 marker (top-left)	F2	☐	☐	☐	☐	☐
U4	USBLC6-2SC6	Pin-1 dot on the ●1 marker	F2	☐	☐	☐	☐	☐
U2	AMS1117-3.3	Big tab = right side	D5	☐	☐	☐	☐	☐
Q1	MMBT3904	Fits one way (3 pads)	F3	☐	☐	☐	☐	☐
Q2	MMBT3904	Fits one way (3 pads)	F3	☐	☐	☐	☐	☐
Q901	AO3400A	Fits one way (3 pads)	G5	☐	☐	☐	☐	☐
D1	SMAJ5.0A	Band on K (toward +5V)	E4	☐	☐	☐	☐	☐
D901	SS34	Band on K (toward V+)	G5	☐	☐	☐	☐	☐
LED1	LED red	Cathode on K	D4	☐	☐	☐	☐	☐
LED2	LED yellow	Cathode on K	E3	☐	☐	☐	☐	☐
C1	100 uF 10 V	Long leg + in square pad	F4	☐	☐	☐	☐	☐
C4	100 uF 10 V	Long leg + in square pad	E5	☐	☐	☐	☐	☐
U1	ESP-32S (Ai-Thinker)	Antenna over top edge, pin 1 top-left	D1	☐	☐	☐	☐	☐

Solder joints and cleanliness

AREA	CHECK	B1	B2	B3	B4	B5
U3 CH340C	No bridges between the 16 pins; all pins wetted (look from both sides)	☐	☐	☐	☐	☐
U4 USBLC6	No bridges; pin 2 (GND) and pin 5 (+5V) joined to their vias	☐	☐	☐	☐	☐
U1 module	All 38 castellations wetted; no bridge on the left, right or bottom row	☐	☐	☐	☐	☐
U2 regulator	Tab fully soldered to the copper area; three small pins wetted	☐	☐	☐	☐	☐
0805 parts	No tombstones, no missing parts in the tables, fillets on both ends	☐	☐	☐	☐	☐
Empty footprints	Dashed option footprints are still empty and free of solder blobs	☐	☐	☐	☐	☐
Solder jumpers	JP101–JP801 untouched (copper trace still joins the factory pads); JP901 open	☐	☐	☐	☐	☐
Cleaning	Flux removed with isopropyl alcohol; board dry	☐	☐	☐	☐	☐
Loose solder	No solder balls or wire clippings on the board (shake it, look under the module)	☐	☐	☐	☐	☐

Tests (1/2): resistance and voltages

Write the reading for each board in its column. Stop at the first wrong value and look for the cause before going on.

1. Resistance, power off

CHECK	PROBE	EXPECTED	B1	B2	B3	B4	B5
+5V to GND	C1 square pad (+) → any port G	Starts low, climbs (capacitors charging); must end above ~1 kΩ. Under 50 Ω = short.					
+3V3 to GND	C4 square pad (+) → any port G	Same: climbs to above ~1 kΩ.					
+5V to +3V3	C1 + → C4 +	Not a short (above ~1 kΩ).					
VBUS to +5V	J1 pin 5V → C1 +	Under 1 Ω (through the PTC F1).					

2. First power-up

Power from the USB-C breakout with an A-to-C cable. A USB power meter in line is useful: note the current. If anything gets hot or the power LED does not light, unplug at once.

CHECK	PROBE	EXPECTED	B1	B2	B3	B4	B5
VBUS (USB in)	J1 pin 5V → G	4.75–5.25 V					
+5V	C1 + → G	4.7–5.2 V (a little under VBUS)					
+3V3	C4 + → G	3.23–3.37 V					
Power LED	LED1 (red)	On					
Regulator	Touch U2 after 1 minute	Warm at most, never too hot to touch					
EN	R1 pad on the EN side → G	3.3 V (RESET not pressed)					
Port VP	Pin V of P1–P6 and I2C → G	3.3 V					
UART VP	Pin V of UART → G	5 V					
I2C idle	DA and CL pins → G	3.3 V (pull-ups)					

Tests (2/2): USB, firmware and function

With the board on USB

3. USB, firmware and function

CHECK	HOW	EXPECTED	B1	B2	B3	B4	B5
USB serial	Plug into the PC	CH340 appears: COMx (Windows) / ttyUSB0 (Linux)	☐	☐	☐	☐	☐
Flash	Flash the test config below	Auto-reset works; if not, hold BOOT, tap RESET	☐	☐	☐	☐	☐
Status LED	After boot	LED2 (yellow) blinks while Wi-Fi is not connected	☐	☐	☐	☐	☐
Wi-Fi	Connect to AP "modbase-test", open 192.168.4.1	Web page with the switches	☐	☐	☐	☐	☐
BOOT button	Press SW2, watch the web page	BOOT button turns ON	☐	☐	☐	☐	☐
RESET	Press SW1	Board restarts (LED2 blinks again)	☐	☐	☐	☐	☐
Port A pins	Switch each A on, measure pin A → G	3.3 V on, 0 V off (P1–P6)	☐	☐	☐	☐	☐
UART TX	Switch UART TX, measure TX → G	3.3 V / 0 V	☐	☐	☐	☐	☐
OUT	LED + 1 kΩ from UART pin V (5 V) to OUT pin	LED on when OUT switch is on	☐	☐	☐	☐	☐

Test firmware (ESPHome)

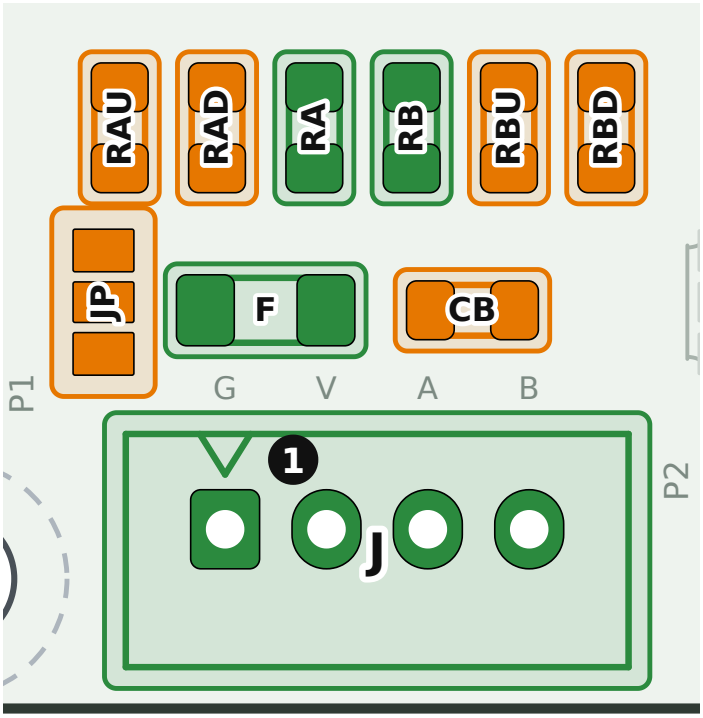
```
esphome:
  name: modbase-test
esp32:
  board: esp32dev
logger:
wifi:
  ap:
    ssid: "modbase-test"
captive_portal:
web_server:
  port: 80
status_led:
  pin: GPIO2
binary_sensor:
  - platform: gpio
    name: "BOOT button"
    pin: {number: GPIO0, inverted: true}
switch:
  - {platform: gpio, name: "P1 A", pin: GPIO25}
  - {platform: gpio, name: "P2 A", pin: GPIO26}
  - {platform: gpio, name: "P3 A", pin: GPIO27}
  - {platform: gpio, name: "P4 A", pin: GPIO13}
  - {platform: gpio, name: "P5 A", pin: GPIO4}
  - {platform: gpio, name: "P6 A", pin: GPIO18}
  - {platform: gpio, name: "UART TX", pin: GPIO17}
  - {platform: gpio, name: "OUT", pin: GPIO23}
```

Notes

- Flash over USB: `esphome run modbase-test.yaml` and pick the CH340 port.
- ESPHome warns that GPIO0 and GPIO2 are strapping pins: expected, they are the BOOT button and status LED.
- The A pins are measured at the connector, through the 330 R: with nothing connected you read the full 3.3 V.
- B pins are inputs; test them when you configure a port (NTC, pulse, divider).
- IO36/IO39 (P1/P2 B) cannot drive anything: they are input-only.
- When the board passes, flash your real configuration over USB or OTA.
- Mains: this board is low voltage only. Keep relay modules and meters behind the divider in the box, and flash with mains disconnected.

Port options

Every port has the same layout. Port *n* uses references with that number: P1 = R101…R106, C101, F101, JP101. Green = fitted in step 4, orange = empty until you need it. Drawing: P1 at the bottom-left corner of the board.



POS.	REF	DEFAULT	JOB
RA	Rn01	330 R fitted	Series resistor, connector A ↔ GPIO
RAU	Rn02	empty	Pull-up, A to 3V3 (connector side)
RAD	Rn03	empty	Pull-down, A to GND
RB	Rn04	1 k fitted	Series resistor, connector B ↔ GPIO
RBU	Rn05	empty	Pull-up, B to 3V3 (connector side)
RBD	Rn06	empty	B to GND on the GPIO side (divider bottom)
CB	Cn01	empty	Filter capacitor, B to GND on the GPIO side
F	Fn01	0.5 A PTC	Protects VP
JP	JPn01	trace 3V3	VP source jumper

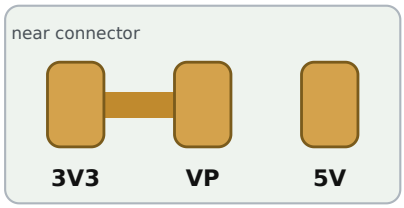
PORT	REFS	A	B	VP
P1	101–106	IO25	IO36	3V3
P2	201–206	IO26	IO39	3V3
P3	301–306	IO27	IO34	3V3
P4	401–406	IO13	IO35	3V3
P5	501–506	IO4	IO32	3V3
P6	601–606	IO18	IO33	3V3
UART	701–706	IO17	IO16	5V
I2C	801–806	IO21	IO22	3V3

What to fit for each job

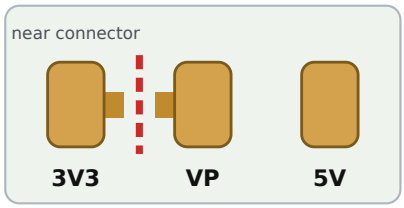
JOB	PIN	VP	FIT (ON TOP OF DEFAULTS)	EXAMPLE ON P3
Relay module, low-level trigger	A	3V3	Nothing	`Defaults only`
Relay module, high-level trigger	A	3V3	RAD 10 k	`R303 = 10k`
DS18B20 (1-Wire)	A	3V3	RAU 4.7 k	`R302 = 4k7`
Button or dry contact	A	-	Nothing (internal pull-up); RAU 10 k for long cables	`R302 = 10k (long cable)`
NTC 10 k (B3950)	B	-	RBU 10 k 1 %, CB 100 nF	`R305 = 10k 1%, C301 = 100nF`
0-3.3 V analog sensor	B	as sensor needs	CB 100 nF	`C301 = 100nF`
5 V logic or pulse (flow sensor with 5 V output)	B (P3-P6)	5V	RB 10 k (replace 1 k), RBD 22 k, CB 1 nF	`R304 = 10k, R306 = 22k, C301 = 1nF`
Open-collector pulse (flow sensor)	A	5V	RAU 10 k	`R302 = 10k`
PZEM-004T energy meter	UART port	5V (default)	Nothing	`J701`
RS485 module (Modbus)	UART port + P5 or P6 B	5V	Remove R706 if the module is 3.3 V	`Direction pin on IO32 (P5 B)`
I2C sensors (BME280, SHT3x, ADS1115, OLED)	I2C port	3V3 (default)	Nothing	`J801`
Solid-state relay (DC input), buzzer, 12/24 V LED, fan	OUT	V+ from load supply	Nothing	`J901`

VP and OUT jumpers

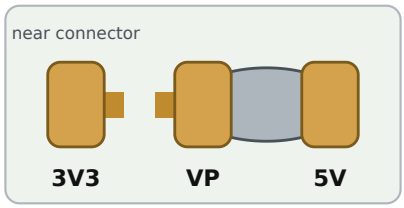
Each port's supply pin **VP** comes from a three-pad solder jumper. The pad **nearest the connector is 3V3**, the far pad is 5V, the middle pad is VP. The board ships with a thin copper trace joining the default pair: 3V3 on P1–P6 and I2C, **5V on UART**.



1. As made: 3V3 joined by a trace



2. Cut the trace between the pads



3. Bridge VP to 5V with solder

Changing a port to 5V

1. Power off. Find the jumper (table below), between the connector and the PTC.
2. Cut the trace between the 3V3 pad and the middle pad with a sharp hobby knife: two or three light strokes, across the gap only.
3. Check with the multimeter: **no continuity** between the two pads.
4. Add flux and bridge the middle pad to the 5V pad with a blob of solder.
5. Check: VP pin of the connector reads 5 V after power-up. Write the change in the table.
6. To go back: remove the blob with wick, bridge the 3V3 side with solder instead (the trace is gone).

Only switch a port to 5V for a device that needs it, and only if the device's signal into the ESP32 stays at 3.3 V (or the port has the 10 k / 22 k divider on B). The ESP32 pins are not 5 V tolerant.

PORT	JUMPER	DEFAULT	GRID	CHANGED TO	DATE
P1	JP101	3V3	A6		
P2	JP201	3V3	C6		
P3	JP301	3V3	D6		
P4	JP401	3V3	F6		
P5	JP501	3V3	B2		
P6	JP601	3V3	B4		
UART	JP701	5V	G3		
I2C	JP801	3V3	G4		

JP901: OUT supply

JP901 is a two-pad jumper next to the OUT connector, **open** as made. Bridge it only to power a 5 V load from the board (pin V+ then carries +5V). **Never bridge it while an external 12/24 V supply is on V+**: that would feed 12/24 V into the board's 5 V rail.

Board record

One line per board. Keep this page with the boards; it is the record of what was built and how each one tested.

BOARD	ASSEMBLED	INSPECTED	+5V (V)	+3V3 (V)	CURRENT (MA)	USB / FLASH	PORTS OK	CHANGES MADE / NOTES
B1								
B2								
B3								
B4								
B5								

Port configuration per board

Write what is plugged into each port and which option parts you fitted (for example "NTC: R305 10k1%, C301 100n").

BOARD	P1	P2	P3	P4	P5	P6	UART	I2C	OUT
B1									
B2									
B3									
B4									
B5									

Parts index

All 115 references · grid squares as on the drawings

REF	VALUE	PKG	GRID	WHEN
C1	100 uF 10 V	Radial 6.3 mm	F4	step 7
C2	10 uF	0805	D4	step 3
C3	22 uF	0805	E5	step 3
C4	100 uF 10 V	Radial 6.3 mm	E5	step 7
C5	100 nF	0805	C1	step 3
C6	10 uF	0805	C1	step 3
C7	1 uF	0805	C1	step 3
C8	100 nF	0805	F2	step 3
C101	empty	0805	B6	empty (option)
C201	empty	0805	D6	empty (option)
C301	empty	0805	E6	empty (option)
C401	empty	0805	G6	empty (option)
C501	empty	0805	A3	empty (option)
C601	empty	0805	A5	empty (option)
C701	empty	0805	G2	empty (option)
C801	empty	0805	G3	empty (option)
D1	SMAJ5.0A	SMA	E4	step 5
D901	SS34	SMA	G5	step 5
F1	PTC 1.1 A hold	1812	E4	step 5
F101	PTC 0.5 A hold	1206	B6	step 4
F201	PTC 0.5 A hold	1206	C6	step 4
F301	PTC 0.5 A hold	1206	E6	step 4
F401	PTC 0.5 A hold	1206	F6	step 4
F501	PTC 0.5 A hold	1206	A3	step 4
F601	PTC 0.5 A hold	1206	A4	step 4
F701	PTC 0.5 A hold	1206	G2	step 4
F801	PTC 0.5 A hold	1206	G4	step 4
J1	XH 4P socket	XH 4P	F1	step 7
J101	XH 4P socket	XH 4P	B7	step 7
J201	XH 4P socket	XH 4P	C7	step 7
J301	XH 4P socket	XH 4P	E7	step 7
J401	XH 4P socket	XH 4P	F7	step 7
J501	XH 4P socket	XH 4P	A3	step 7
J601	XH 4P socket	XH 4P	A4	step 7
J701	XH 4P socket	XH 4P	H2	step 7
J801	XH 4P socket	XH 4P	H4	step 7
J901	XH 3P socket	XH 3P	H5	step 7
J902	empty	1x5 2.54	C3	empty (option)
JP101	–	Jumper	A6	jumper
JP201	–	Jumper	C6	jumper
JP301	–	Jumper	D6	jumper
JP401	–	Jumper	F6	jumper
JP501	–	Jumper	B2	jumper
JP601	–	Jumper	B4	jumper
JP701	–	Jumper	G3	jumper
JP801	–	Jumper	G4	jumper
JP901	–	Jumper	G5	jumper
LED1	LED red	0805 LED	D4	step 5
LED2	LED yellow	0805 LED	E3	step 5
Q1	MMBT3904	SOT-23	F3	step 2
Q2	MMBT3904	SOT-23	F3	step 2
Q901	AO3400A	SOT-23	G5	step 2
R1	10 k	0805	C1	step 3
R2	10 k	0805	E3	step 3
R3	10 k	0805	F4	step 3
R4	10 k	0805	F4	step 3
R5	470 R	0805	F3	step 3
R10	1 k	0805	D4	step 3
R11	1 k	0805	E3	step 3
R101	330 R	0805	B6	step 4
R102	empty	0805	A6	empty (option)
R103	empty	0805	A6	empty (option)
R104	1 k	0805	B6	step 4
R105	empty	0805	B6	empty (option)
R106	empty	0805	B6	empty (option)
R201	330 R	0805	C6	step 4
R202	empty	0805	C6	empty (option)
R203	empty	0805	C6	empty (option)
R204	1 k	0805	D6	step 4
R205	empty	0805	D6	empty (option)
R206	empty	0805	D6	empty (option)
R301	330 R	0805	E6	step 4
R302	empty	0805	D6	empty (option)
R303	empty	0805	E6	empty (option)
R304	1 k	0805	E6	step 4
R305	empty	0805	E6	empty (option)
R306	empty	0805	F6	empty (option)
R401	330 R	0805	F6	step 4
R402	empty	0805	F6	empty (option)
R403	empty	0805	F6	empty (option)
R404	1 k	0805	G6	step 4
R405	empty	0805	G6	empty (option)
R406	empty	0805	G6	empty (option)
R501	330 R	0805	B3	step 4
R502	empty	0805	B2	empty (option)
R503	empty	0805	B3	empty (option)
R504	1 k	0805	B3	step 4
R505	empty	0805	B3	empty (option)
R506	empty	0805	B4	empty (option)
R601	330 R	0805	B4	step 4
R602	empty	0805	B4	empty (option)
R603	empty	0805	B4	empty (option)
R604	1 k	0805	B5	step 4
R605	empty	0805	B5	empty (option)
R606	empty	0805	B5	empty (option)
R701	330 R	0805	G2	step 4
R702	empty	0805	G3	empty (option)
R703	empty	0805	G2	empty (option)
R704	10 k	0805	G2	step 4
R705	empty	0805	G2	empty (option)
R706	22 k	0805	G2	step 4
R801	100 R	0805	G4	step 4
R802	4.7 k	0805	G4	step 4
R803	empty	0805	G4	empty (option)
R804	100 R	0805	G4	step 4
R805	4.7 k	0805	G3	step 4
R806	empty	0805	G3	empty (option)
R901	100 R	0805	F5	step 3
R902	100 k	0805	F5	step 3
SW1	Tactile switch	6x6 THT	A1	step 7
SW2	Tactile switch	6x6 THT	A1	step 7
U1	ESP-32S (Ai-Thinker)	Module	D1	step 6
U2	AMS1117-3.3	SOT-223	D5	step 2
U3	CH340C	SOIC-16	F2	step 1
U4	USBLC6-2SC6	SOT-23-6	F2	step 1